

SQL Assignment Advanced

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DS68

1. Find the total revenue collected by each shipper.

```
SELECT s.shipper_name, SUM(oi.quantity * oi.price_each) AS total_revenue FROM  
Orders o JOIN Shippers s ON o.shipper_id = s.shipper_id  
JOIN Order_Items oi ON o.order_id = oi.order_id GROUP BY s.shipper_name;
```

	shipper_name	total_revenue
▶	BlueDart Express	149700.00
	DTDC Courier	200200.00
	Delhivery Logistics	122400.00
	Ekart Logistics	123500.00
	Gati Express	105400.00
	India Post Speed	155100.00
	Shadowfax Delivery	100800.00

2. Show the top 5 highest-spending customers along with their total payments.

```
select c.name , sum(oi.quantity * oi.price_each) as total_payment from orders o join  
customers c on o.customer_id = c.customer_id  
join order_items oi on o.order_id = oi.order_id GROUP BY c.name ORDER BY  
total_payment DESC LIMIT 5;
```

	name	total_payment
▶	Rohit Kumar	152000.00
	Megha Kapoor	88500.00
	Sneha Sharma	83100.00
	Ritika Gupta	77600.00
	Tanya Mishra	75200.00

3. Find product categories where the average selling price is greater than 8000.

```
select category, avg(price) as avg_price from products group by category having avg_price > 8000;
```

#3. Find product categories where the average selling price is greater than 8000.

```
select category, avg(price) as avg_price from products group by category having avg_price > 8000;
```

Result Grid			Filter Rows:
	category	avg_price	
▶	Electronics	67749.000000	

4. Show the total number of orders placed per city, sorted by highest to lowest.

```
select c.city , sum(oi.quantity) as quantity_by_city from orders o join customers c on c.customer_id = o.customer_id
```

```
join order_items oi on o.order_id = oi.order_id group by c.city;
```

	city	quantity_by_city
▶	Delhi	19
	Mumbai	8
	Ahmedabad	10
	Hyderabad	6
	Chennai	18
	Pune	6
	Bengaluru	16
	Kolkata	10
	Lucknow	9
	Bhubaneswar	12
	Jaipur	8
	Noida	8
	Indore	11

5. Find suppliers who supply more than 1 product category.

```
select p.category, s.supplier_name, SUM(oi.quantity) AS total_qty from products p join  
suppliers s on p.supplier_id = s.supplier_id
```

```
join order_items oi on oi.product_id = p.product_id group by p.category ,  
s.supplier_name having sum(oi.quantity)>1;
```

	category	supplier_name	total_qty
▶	Electronics	Reliance Digital	16
	Electronics	Croma Electronics	31
	Wearables	Flipkart Seller Hub	8
	Wearables	Amazon India Vendor	13
	Fashion	Lifestyle Stores	37
	Home Appliances	Tata Cliq	15
	Electronics	Vijay Sales	10
	Personal Care	Metro Electronics	11

6. Show each order along with the number of items it contains.

```
select o.order_id, COUNT(oi.product_id) from orders o join order_items oi on oi.order_id  
= o.order_id group by o.order_id;
```

	order_id	COUNT(oi.product_id)
▶	1	1
	2	1
	3	1
	4	1
	5	1
	6	1
	7	1
	8	1
	9	1
	10	1
	11	1
	12	1
	13	1
	14	1
	15	1

7. Find customers who have spent more than the average spending of all customers.

```
SELECT c.name, SUM(oi.price_each * oi.quantity) AS total_spent FROM customers c  
JOIN orders o ON o.customer_id = c.customer_id
```

```
JOIN order_items oi ON oi.order_id = o.order_id GROUP BY c.name HAVING total_spent  
> (SELECT AVG(customer_total)
```

```
FROM (SELECT SUM(oi.price_each * oi.quantity) AS customer_total FROM orders o JOIN  
order_items oi
```

```
ON oi.order_id = o.order_id GROUP BY o.customer_id) AS avg_table);
```

	name	total_spent
▶	Rohit Kumar	152000.00
	Sneha Sharma	83100.00
	Amit Patel	70000.00
	Arjun Mehta	66000.00
	Ritika Gupta	77600.00
	Megha Kapoor	88500.00
	Tanya Mishra	75200.00
	Aditya Jain	72000.00

8. List all products whose price is higher than the average product price in their category.

```
select product_name, price, category from products p1 where price > (select avg(price)  
from products p2 where p1.category=p2.category);
```

	product_name	price	category
▶	iPhone 15	79999.00	Electronics
	Samsung Galaxy S24	74999.00	Electronics
	Noise Smartwatch	4999.00	Wearables
	Running Shoes	2999.00	Fashion

9. Show customers who placed at least one order with total_amount greater than 50,000.

```
select distinct c.customer_id , c.name, sum(oi.price_each * oi.quantity ) as  
total_amount from customers c join orders o on o.customer_id = c.customer_id  
join order_items oi on oi.order_id = o.order_id group by c.customer_id , c.name,  
o.order_id having total_amount > 50000;
```

	customer_id	name	total_amount
▶	1	Rohit Kumar	66000.00

10. Find customers who placed more orders than the average number of orders per customer.

```
SELECT c.customer_id, c.name, COUNT(o.order_id) AS total_orders FROM customers c  
JOIN orders o ON o.customer_id = c.customer_id
```

```
GROUP BY c.customer_id, c.name HAVING COUNT(o.order_id) > (SELECT  
AVG(order_count)FROM (SELECT COUNT(order_id) AS order_count
```

```
FROM orders GROUP BY customer_id) t);
```

	customer_id	name	total_orders
▶	2	Sneha Sharma	6
	7	Arjun Mehta	6

11. Find the most expensive product(s) in the catalog.

```
select product_name, price from products order by price desc limit 1;
```

	product_name	price
▶	iPhone 15	79999.00

12. Rank customers based on their total spending.

```
select c.customer_id , c.name, sum(oi.price_each * oi.quantity ) as total_spending,  
rank() over(order by sum(oi.price_each * quantity)desc) as rank1
```

```
from customers c join orders o on o.customer_id = c.customer_id join order_items oi on  
oi.order_id = o.order_id
```

group by c.customer_id , c.name;

	customer_id	name	total_spending	rank1
▶	1	Rohit Kumar	152000.00	1
	12	Megha Kapoor	88500.00	2
	2	Sneha Sharma	83100.00	3
	8	Ritika Gupta	77600.00	4
	14	Tanya Mishra	75200.00	5
	15	Aditva Jain	72000.00	6

13. Find cumulative sales amount by order date.

```
SELECT o.order_date, SUM(oi.quantity * oi.price_each) AS daily_sales,  
SUM(SUM(oi.quantity * oi.price_each)) OVER (ORDER BY o.order_date) AS  
cumulative_sales
```

```
FROM Orders o JOIN Order_Items oi ON o.order_id = oi.order_id GROUP BY  
o.order_date;
```

	order_date	daily_sales	cumulative_sales
▶	2024-01-05	30000.00	30000.00
	2024-01-06	12000.00	42000.00
	2024-01-07	7500.00	49500.00
	2024-01-08	25600.00	75100.00
	2024-01-09	14000.00	89100.00
	2024-01-10	7000.00	96100.00
	2024-01-11	10600.00	106700.00
	2024-01-12	22000.00	128700.00
	2024-01-13	30000.00	158700.00
	2024-01-14	25800.00	184500.00

14. Get each customer's order count and show their percentage contribution.

```
SELECT customer_id, COUNT(order_id) AS total_orders, COUNT(order_id) * 100.0 /  
SUM(COUNT(order_id))  
OVER () AS percentage FROM Orders GROUP BY customer_id;
```

	customer_id	total_orders	percentage
▶	1	5	6.66667
	2	6	8.00000
	3	5	6.66667
	4	4	5.33333
	5	5	6.66667
	6	4	5.33333
	7	6	8.00000
	8	5	6.66667

15. Show the most recent order per customer.

```
SELECT * FROM ( SELECT *,ROW_NUMBER() OVER (PARTITION BY customer_id ORDER
BY order_date DESC) AS rn
```

```
FROM Orders) t WHERE rn = 1;
```

	order_id	customer_id	order_date	payment_id	shipper_id	rn
▶	71	1	2024-02-21	2	7	1
	72	2	2024-02-21	3	6	1
	73	3	2024-02-22	4	1	1
	59	4	2024-02-13	2	5	1
	74	5	2024-02-23	6	5	1
	52	6	2024-02-09	3	7	1
	75	7	2024-02-23	7	4	1
	63	8	2024-02-15	5	2	1
	64	9	2024-02-16	4	7	1
	65	10	2024-02-16	2	6	1
	66	11	2024-02-17	3	1	1

16. List each product with its sales quantity and rank products within each category by total sales.

```
SELECT p.product_name, p.category, SUM(oi.quantity) AS total_sales, RANK() OVER
(PARTITION BY p.category ORDER BY SUM(oi.quantity) DESC)
```

```
AS rank_no FROM Products p JOIN Order_Items oi ON p.product_id = oi.product_id
```

```
GROUP BY p.product_name, p.category;
```

	product_name	category	total_sales	rank_no
▶	Lenovo Laptop	Electronics	18	1
	iPhone 15	Electronics	16	2
	Samsung Galaxy S24	Electronics	13	3
	Sony Bravia 55" TV	Electronics	10	4
	Running Shoes	Fashion	22	1
	Kurta Set	Fashion	15	2
	Prestige Mixer Grinder	Home Appliances	15	1
	Philips Trimmer	Personal Care	11	1
	Boat Earbuds	Wearables	13	1
	Noise Smartwatch	Wearables	8	2

17. Categorize products as 'High', 'Medium', or 'Low' price using CASE.

select product_name , case when price >60000 then "High" when price between 5000 and 50000 then "Medium" else "low" end as Price_category from products;

	product_name	Price_category
▶	iPhone 15	High
	Samsung Galaxy S24	High
	Noise Smartwatch	low
	Boat Earbuds	low
	Kurta Set	low
	Running Shoes	low
	Prestige Mixer Grinder	Medium
	Sony Bravia 55" TV	low
	Lenovo Laptop	low
	Philips Trimmer	low

18. Use a CTE to find top 3 customers by spending.

WITH customer_spending AS (SELECT c.customer_id, c.name,SUM(oi.quantity * oi.price_each) AS total_spent

FROM Customers c JOIN Orders o ON c.customer_id = o.customer_id JOIN Order_Items oi ON o.order_id = oi.order_id

GROUP BY c.customer_id, c.name) SELECT * FROM customer_spending

ORDER BY total_spent DESC LIMIT 3;

	customer_id	name	total_spent
▶	1	Rohit Kumar	152000.00
	12	Megha Kapoor	88500.00
	2	Sneha Sharma	83100.00

19. Use a CTE with CASE to classify customers by loyalty (based on number of orders).

```
WITH order_counts AS ( SELECT customer_id, COUNT(order_id) AS total_orders
FROM Orders GROUP BY customer_id)SELECT customer_id,
CASE WHEN total_orders >= 5 THEN 'High' WHEN total_orders >= 3 THEN 'Medium'
ELSE 'Low' END AS loyalty FROM order_counts;
```

	customer_id	loyalty
▶	1	High
	2	High
	3	High
	4	Medium
	5	High
	6	Medium
	7	High
	8	High
	9	High
	10	High

20. Find monthly revenue growth percentage compared to the previous month.

```
WITH monthly_sales AS (SELECT DATE_FORMAT(o.order_date, '%Y-%m') AS month,
SUM(oi.quantity * oi.price_each) AS revenue from Orders o join Order_Items oi on
o.order_id = oi.order_id group by month)
SELECT month, revenue, (revenue) OVER (ORDER BY month) AS prev_month,
((revenue – lag (revenue) OVER (ORDER BY month)) * 100.0 /
Lag (revenue) OVER (ORDER BY month)) AS growth_percentage
from monthly_sales;
```

	month	revenue	prev_month	growth_percentage
▶	2024-01	479300.00	NULL	NULL
	2024-02	477800.00	479300.00	-0.3129564

21. Find top 2 customers per city based on total spending.

WITH customer_spending AS (SELECT c.city, c.name, SUM(oi.quantity * oi.price_each)
AS total_spent

FROM Customers c JOIN Orders o ON c.customer_id = o.customer_id JOIN Order_Items
oi ON o.order_id = oi.order_id

GROUP BY c.city, c.name) SELECT * FROM (SELECT *,

RANK() OVER (PARTITION BY city ORDER BY total_spent DESC) AS rnk FROM
customer_spending) t

WHERE rnk <= 2;

	city	name	total_spent	rnk
▶	Ahmedabad	Amit Patel	70000.00	1
	Bengaluru	Arjun Mehta	66000.00	1
	Bhubaneswar	Ananya Das	30400.00	1
	Chennai	Suresh Iyer	48400.00	1
	Chennai	Karan Singh	24800.00	2
	Delhi	Rohit Kumar	152000.00	1
	Delhi	Ravi Shankar	55500.00	2
	Hyderabad	Priya Reddy	42600.00	1
	Indore	Aditya Jain	72000.00	1
	Jaipur	Megha Kapoor	88500.00	1
	Kolkata	Ritika Gupta	77600.00	1
	Lucknow	Vikram Joshi	24600.00	1
	Mumbai	Sneha Sharma	83100.00	1
	Noida	Tanya Mishra	75200.00	1
	Pune	Neha Verma	46400.00	1

22. Find top 3 cities with highest sales revenue including shipper name.

SELECT c.city, s.shipper_name, SUM(oi.quantity * oi.price_each) AS revenue FROM
Customers c

JOIN Orders o ON c.customer_id = o.customer_id JOIN Shippers s ON o.shipper_id = s.shipper_id

JOIN Order_Items oi ON o.order_id = oi.order_id GROUP BY c.city, s.shipper_name
ORDER BY revenue DESC LIMIT 3;

	city	shipper_name	revenue
▶	Delhi	DTDC Courier	66000.00
	Noida	Delhivery Logistics	56000.00
	Jaipur	BlueDart Express	55000.00

23. List all orders with customer name, product name, supplier, and shipper.

SELECT o.order_id, c.name, p.product_name, s.supplier_name, sh.shipper_name
FROM Orders o

JOIN Customers c ON o.customer_id = c.customer_id

JOIN Order_Items oi ON o.order_id = oi.order_id

JOIN Products p ON oi.product_id = p.product_id

JOIN Suppliers s ON p.supplier_id = s.supplier_id

JOIN Shippers sh ON o.shipper_id = sh.shipper_id;

	order_id	name	product_name	supplier_name	shipper_name
▶	1	Rohit Kumar	iPhone 15	Reliance Digital	Delhivery Logistics
	12	Megha Kapoor	iPhone 15	Reliance Digital	India Post Speed
	21	Ritika Gupta	iPhone 15	Reliance Digital	Ekart Logistics
	31	Ritika Gupta	iPhone 15	Reliance Digital	DTDC Courier
	41	Rohit Kumar	iPhone 15	Reliance Digital	India Post Speed
	50	Ananya Das	iPhone 15	Reliance Digital	DTDC Courier
	61	Amit Patel	iPhone 15	Reliance Digital	BlueDart Express
	70	Aditya Jain	iPhone 15	Reliance Digital	DTDC Courier
	4	Priya Reddy	Samsung Galaxy S24	Croma Electronics	BlueDart Express
	11	Suresh Iyer	Samsung Galaxy S24	Croma Electronics	BlueDart Express
	22	Rohit Kumar	Samsung Galaxy S24	Croma Electronics	DTDC Courier
	32	Arjun Mehta	Samsung Galaxy S24	Croma Electronics	Gati Express
	40	Megha Kapoor	Samsung Galaxy S24	Croma Electronics	BlueDart Express
	52	Neha Verma	Samsung Galaxy S24	Croma Electronics	Shadowfax Delivery
	60	Sneha Sharma	Samsung Galaxy S24	Croma Electronics	India Post Speed
	69	Tanva Mishra	Samsung Galaxy S24	Croma Electronics	Delhivery Logistics

24. Show total sales per supplier along with average order value.

```
SELECT s.supplier_name, SUM(oi.quantity * oi.price_each) AS total_sales,  
AVG(oi.quantity * oi.price_each) AS avg_order_value FROM Suppliers s  
JOIN Products p ON s.supplier_id = p.supplier_id  
JOIN Order_Items oi ON p.product_id = oi.product_id  
GROUP BY s.supplier_name;
```

	supplier_name	total_sales	avg_order_value
▶	Reliance Digital	240000.00	30000.000000
	Croma Electronics	307600.00	20506.666667
	Flipkart Seller Hub	96000.00	13714.285714
	Amazon India Vendor	104000.00	13000.000000
	Tata Cliq	27000.00	3375.000000
	Vijay Sales	35000.00	5000.000000
	Lifestyle Stores	70500.00	4406.250000
	Metro Electronics	77000.00	12833.333333

25. Show product categories that contributed more than 30% of total sales revenue.

```
WITH category_sales AS ( SELECT p.category, SUM(oi.quantity * oi.price_each) AS  
revenue  
FROM Products p JOIN Order_Items oi ON p.product_id = oi.product_id GROUP BY  
p.category),  
total_sales AS (SELECT SUM(revenue) AS total FROM category_sales)  
SELECT c.category, c.revenue FROM category_sales c, total_sales t  
WHERE (c.revenue * 100.0 / t.total) > 30;
```

	category	revenue
▶	Electronics	582600.00